1.Overview

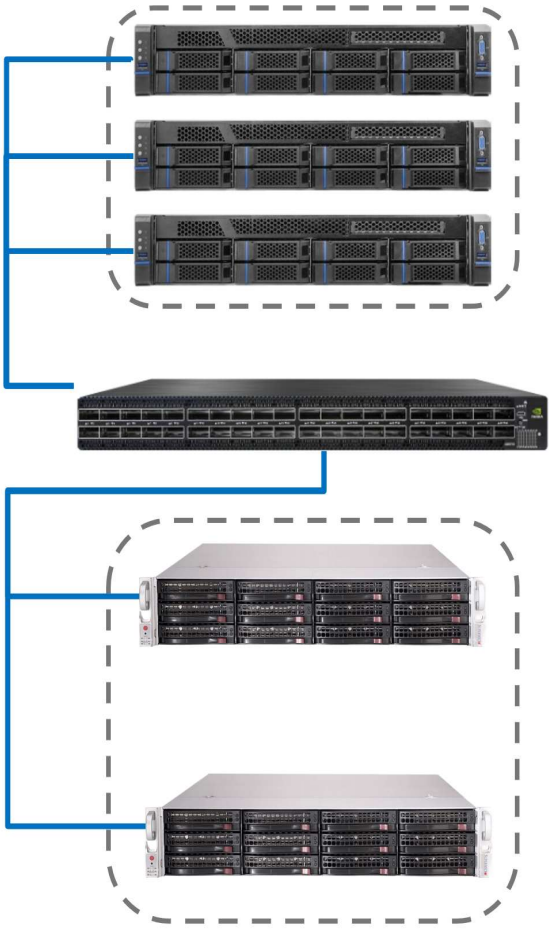
This benchmark test uses a **two-node storage server cluster** and **three client nodes**, tested by TTA (Telecommunications Technology Association) targeting Gluesys storage systems ExaStor. The clients and storage nodes are configured as follows:

Configuration Summary

System name of Submission	Submitter Name	Submission Date
ExaStor (Gluesys)	Byoungjun Seo	2025. 07. 11
Version of the Benchmark	Solution type of the Submission	Submission Division
MLPerf Storage v2.0	On-Premise Appliance	CLOSED

2.Hardware Diagram

System Under Test Configuration Overview
(System Topology)



Client Node

3 x KTNF KR580S2

- 2 x Intel Xeon Platinum 8380 2.30GHz
- 16 x 64GB (1TB) Memory
- 1 x 200Gb 1-Port Infiniband Adaptor

200Gb Infiniband Switch

Storage Node

2 x Gluesys ExaStor

Storage1

- 2 x Intel Xeon Gold 6330 2.00GHz
- 8 x 16GB (128GB) Memory
- 1 x NVMe Optane SSD 280GB
- 8 x NVMe NAND SSD 3.84TB
- 4 x NVMe NAND SSD 7.68TB

Storage2

- 2 x Intel Xeon Gold 6330 2.00GHz
- 8 x 16GB (128GB) Memory
- 8 x NVMe NAND SSD 3.84TB
- 4 x NVMe NAND SSD 7.68TB

2.1 Client Server

This client is equipped with a two Intel 8380 CPU, 1TB of memory, and 1 200Gb IB network cards.

Component	Description
CPU Model	Intel Platinum 8380
CPU Count	2 CPUs per client
Memory	1TB DDR4 per client
NIC Model	MT28908
NIC Details	1 port / 200GB/s per NIC
NIC Count	1 NIC per client
Rack Units	2U
Design Power	800W

Note: *All three client nodes are identical in configuration.*

2.2 Storage Server

The storage server uses the Gluesys System ExaStor storage server, equipped with two Intel 6330, 128GB of memory, and 1 200Gb IB network cards.

Storage1

Component	Description
CPU Model	Intel Gold 6330
CPU Count	2 CPUs per client
Memory	128GB DDR4 per storage
NVMe Optane SSD	1 x 280GB
NVMe NAND SSD	(8 x 3.84TB) + (4 x 7.68TB)
NIC Model	MT28908
NIC Details	1 port / 200GB/s per NIC
NIC Count	1 NIC per storage
Rack Units	2U
Design Power	1200W

Storage2

Component	Description

CPU Model	Intel Gold 6330
CPU Count	2 CPUs per client
Memory	128GB DDR4 per storage
NVMe NAND SSD	(8 x 3.84TB) + (4 x 7.68TB)
NIC Model	MT28908
NIC Details	1 port / 200GB/s per NIC
NIC Count	1 NIC per storage
Rack Units	2U
Design Power	1200W

Note: *Storage1 has an additional Optane SSD (NVMe Optain SSD) not present in Storage2.*

2.3 Switch

NVIDIA QM8700 Switch

3.Software

3.1 Client Server

Client OS - Rocky Linux 8.10 (Green Obsidian)

Client Kernel - 4.18.0-553.el8_10.x86_64

RDMA Core – 48.0-1.el8

libibverbs – 48.0-1.el8

NIC Driver (mlx5_core) – 4.18.0-553.el8_10.x86_64 (OS inbox)

Lustre Client – 2.15.5

3.2 Storage Server

Client OS - Rocky Linux 8.10 (Green Obsidian)

Client Kernel - 4.18.0-553.el8_10.x86_64

RDMA Core – 48.0-1.el8

libibverbs – 48.0-1.el8

NIC Driver (mlx5_core) – 4.18.0-553.el8_10.x86_64 (OS inbox)

Lustre File System – 2.15.7

4.Settings

4.1 Client Server

Example Lustre mount command from client:

```
mount -t lustre 10.102.9.135@o2ib:/cvol /mnt
```

4.2 Storage Server

Example command to display Lustre MDT, MGT, and OST configuration:

```
(base) [root@mlpstorage-driver1 ~]# lctl dl
  0 UP mgc MGC10.102.9.135@o2ib 27be2541-1031-4e72-9073-2ce6e45bebe9 4
  1 UP lov cvol-clilov-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 3
  2 UP lmv cvol-clilmv-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
  3 UP mdc cvol-MDT0000-mdc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
  4 UP mdc cvol-MDT0001-mdc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
  5 UP mdc cvol-MDT0002-mdc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
  6 UP osc cvol-OST0000-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
  7 UP osc cvol-OST0001-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
  8 UP osc cvol-OST0002-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
  9 UP osc cvol-OST0003-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 10 UP osc cvol-OST0004-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 11 UP osc cvol-OST0005-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 12 UP osc cvol-OST0006-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 13 UP osc cvol-OST0007-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 14 UP osc cvol-OST0008-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 15 UP osc cvol-OST0009-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 16 UP osc cvol-OST000a-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 17 UP osc cvol-OST000b-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 18 UP osc cvol-OST000c-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 19 UP osc cvol-OST000d-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 20 UP osc cvol-OST000e-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 21 UP osc cvol-OST000f-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 22 UP osc cvol-OST0010-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 23 UP osc cvol-OST0011-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 24 UP osc cvol-OST0012-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 25 UP osc cvol-OST0013-osc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
 26 UP mdc cvol-MDT0003-mdc-ff2000415d975000 bd08b2c0-e88b-44d1-a218-d8e454090d6b 4
```

5. System Power & Rack Unit Requirements

System component	Power supply unit	Nameplate rated power	Design power	Rack Unit
Client 1	PSU 1(Active)	800 watts	800 watts	2U
	PSU 2(Passive)	800 watts		
Client2	PSU 1(Active)	800 watts	800 watts	2U
	PSU 2(Passive)	800 watts		
Client3	PSU 1(Active)	800 watts	800 watts	2U
	PSU 2(Passive)	800 watts		
Storage 1	PSU 1(Active)	1200 watts	1200 watts	2U
	PSU 2(Passive)	1200 watts		
Storage 2	PSU 1(Active)	1200 watts	1200 watts	2U
	PSU 2(Passive)	1200 watts		
Network switch 1	PSU 1(Active)	1100 watts	1100 watts	1U
	PSU 2(Passive)	1100 watts		
Storage-only System Totals		**4800 watts**	**2400 watts**	**4U**
Full System Totals		**11800 watts**	**5900 watts**	**11U**

Note: *The reported storage-only design power includes only the storage nodes, which are essential to the operation of the storage system. The IB switch and client nodes are excluded from the storage-only totals but are included in the full system configuration table.*